

REAL-TIME DETECTION METHOD FOR CRACKED EGGS BASED ON IMPROVED YOLOV8 AND DUAL VERIFICATION WITH TRIPLE CLASSIFICATION GRANULARITY

CSQ413 Seminar Presentation

Submitted by:

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Outline

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Introduction

- **Commercial Significance:** Eggshell integrity is a vital quality indicator in the poultry industry.
- **High Breakage Rate:** Around 3% of eggs suffer shell damage during production, packing and transit.
- **Financial and Health Risks:** Cracks cause bacterial invasion, rapid spoilage, yolk leakage and financial losses.
- **Current Industry Reliance:** Still primarily rely on manual visual inspection.
- **Manual Bottlenecks:** Labor-intensive, low efficiency and prone to missed defects from worker fatigue.

Limitations of Existing Methods

- **Acoustic Resonance Methods:**

- Analyzes acoustic response from mechanical tapping.
- Vulnerable to severe ambient noise on factory floors; high equipment cost.

- **Traditional Computer Vision:**

- Employs manual feature extraction (Otsu thresholding, morphological filters).
- Slow processing speed; fragile under fluctuating factory illumination.

- **Existing Deep Learning Approaches:**

- High computational complexity; inference speeds struggle to sustain ≥ 30 FPS.
- Evaluated primarily on static, pre-washed egg images.
- Absence of egg rotation creates uninspected blind spots on the bottom shell.

Research Objectives and Key Contributions

Primary Objective: Develop an industrial-grade, real-time online detection method for unwashed cracked eggs on dynamic production lines.

Core Contributions of the Study:

- **Triple Granularity and Dual Verification:** Classifies intact eggs, cracked eggs and crack regions with a joint decision rule to eliminate missed defects.
- **Structural YOLOv8 Optimization:** Embeds DCNv2, BiFPN, WIoUv2 loss and an extra small object head to capture subtle hairline cracks.
- **Live Industrial Deployment:** Deployed on a Kyowa/FP-600 egg conveyor with rotating rollers, achieving 45 FPS and meeting industrial reliability standards.

Literature Survey: Traditional and Two-Stage Methods

Table: Literature survey of traditional image processing and two-stage deep learning methods

Paper, Author, Year	Methodology	Dataset	Result	Limitations
Integrated egg quality system <i>Liang et al. (2020)</i>	Histogram thresholding + Otsu segmentation	Static egg images (lab setup)	98.18% accuracy	Manual feature extraction; sensitive to lighting; static only.
Damage detection on eggshells <i>Datta et al. (2019)</i>	Faster R-CNN (two-stage detector)	Chicken eggshells (static)	75% mAP (severe cracks)	High computational complexity; inference too slow for real-time lines.
Defective egg detection via CNN-BiLSTM <i>Turkoglu (2021)</i>	Hybrid CNN-BiLSTM deep network	Egg surface defect dataset	96.3% accuracy	Heavy network architecture; serial processing limits line throughput.

Literature Survey: Deep Learning and YOLO Adaptations

Table: Literature survey of convolutional and YOLO-based egg crack inspection methods

Paper, Author, Year	Methodology	Dataset	Result	Limitations
Eggshell crack detection <i>Botta et al. (2022)</i>	Deep Convolutional Neural Networks	Static chicken egg images	95.38% accuracy	Optimized only for static scenes; misses bottom cracks due to lack of rotation.
Lightweight duck egg crack detection <i>Yin et al. (2024)</i>	YOLOv5 + Ghost modules + ECA attention	Cracked duck egg dataset	93.8% accuracy	Tested only on duck eggs; binary labeling misses subtle hairline cracks.
Damage detection of unwashed eggs <i>Huang et al. (2023)</i>	YOLOv5 + BiFPN + GSConv + CA attention	Dynamic video of unwashed eggs	96.4% accuracy	Lacks dual verification of localized crack regions; bottom cracks uninspected.

System Hardware and Conveying Mechanism

- **Multi-Channel Conveyor:** 6 parallel channels carrying 60 eggs/min each (total throughput: 360 eggs/min).
- **Compound Motion (Translation + Rotation):** Motor-driven rollers rotate eggs 360 degrees, presenting all shell surfaces to the camera.
- **Parallel Camera Alignment:** CCD camera imaging plane is positioned parallel to egg motion plane to prevent perspective distortion.
- **Regulated Lighting:** Integrated LED illumination box ensures uniform brightness across all channels.

Conveying Device and Vision System Setup

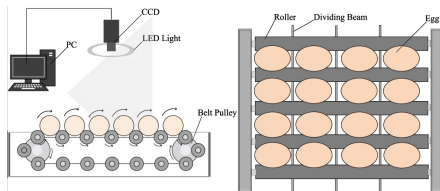
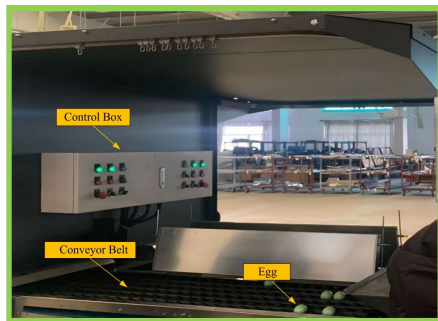


Figure: Left: Egg conveying device with 6 channels (Figure 1). Right: Machine vision system schematic (Figure 2).

Dataset Acquisition and Preparation

- **Industrial Collection:** Captured on live conveyor at Guangzhou Suiping Enterprise Co., Ltd.
- **Real-World Conditions:** Tested on unwashed eggs with dust, stains and factory lighting variations.
- **Balanced Sampling:** 628 raw dynamic images with deliberate inclusion of cracked eggs.
- **Data Augmentation:** Random cropping and flipping (50% probability) expanded training set to 2,972 images.
- **Final Dataset (6:2:2 Split):** 3,222 total images containing 7,611 intact eggs, 2,711 cracked eggs and 2,935 cracked areas.

Triple Classification Granularity

- **Category 1 (Intact Egg):** Normal undamaged egg body.
- **Category 2 (Cracked Egg):** Entire egg body displaying cracks.
- **Category 3 (Crack Region):** Specific localized micro-damage zone.

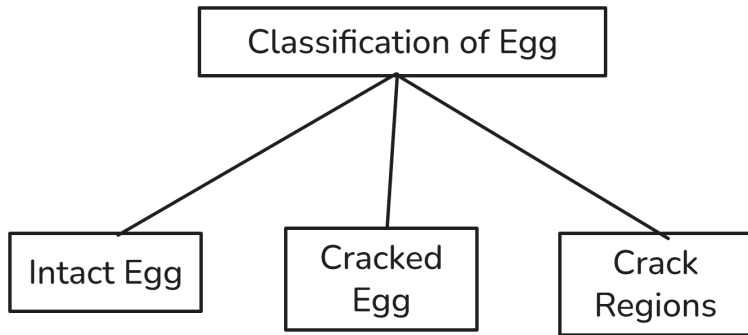


Figure: Classification categories: Intact eggs, cracked eggs and localized crack regions.

Improved YOLOv8 Network Architecture: Main Pipeline

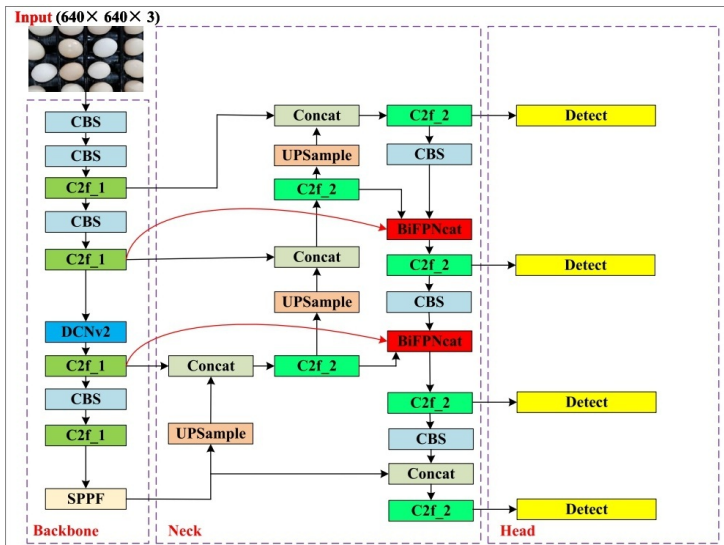


Figure: Improved YOLOv8 overall detection architecture (Source: Base Paper Figure 3).

Improved YOLOv8: Structural Sub-Module Details

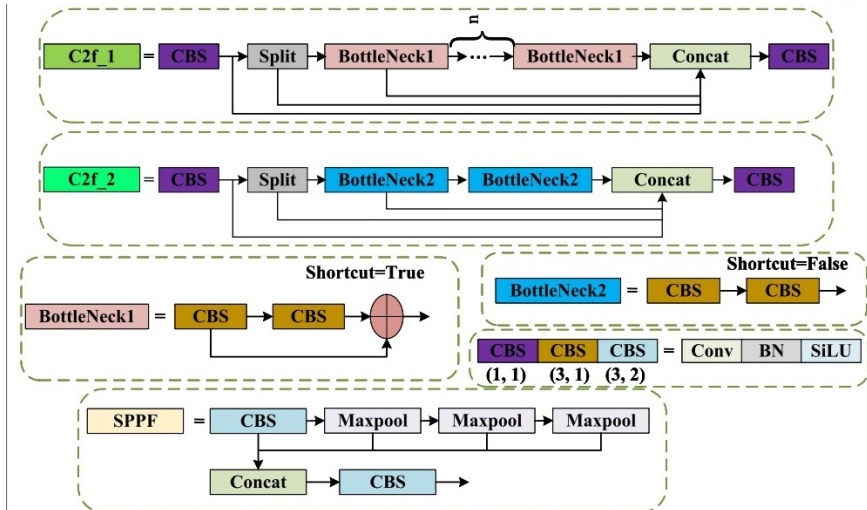


Figure: Detailed network sub-module structures: C2f, BottleNeck and SPPF (Source: Base Paper Figure 3).

Architectural Enhancement 1: Deformable Convolutions (DCNv2)

- **Motivation:** Standard square kernels cannot conform to irregular, jagged crack geometry.
- **Mechanism:** DCNv2 adds learnable spatial offsets and dynamic modulation scalars to adapt receptive fields.

$$Y(q) = \sum_{k=1}^K w_k X(q + q_k + \Delta q_k) \Delta m_k$$

Symbol Definitions:

- X, Y : Input and output feature maps.
- q : Center target coordinate; q_k : Predefined grid offset ($k = 1, \dots, K$).
- w_k : Convolutional kernel weight for the k -th location.
- Δq_k : Learnable 2D spatial offset adapting to geometric crack contours.
- Δm_k : Learnable modulation scalar ($\Delta m_k \in [0, 1]$) scaling feature amplitude.

DCNv2: Sampling Strategy Comparison

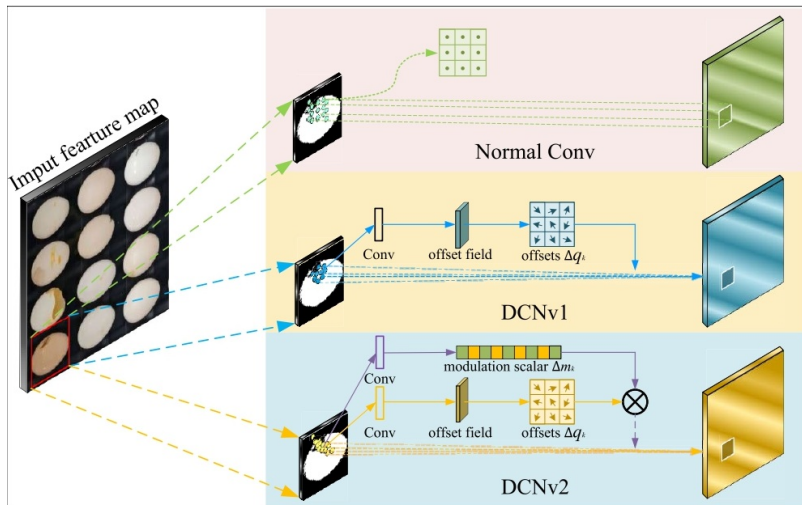


Figure: Sampling comparison of Normal Conv, DCNv1 and DCNv2 (Source: Base Paper Figure 4).

Architectural Enhancement 2: BiFPN Multi-Scale Fusion

- **FPN vs PANet vs BiFPN Evolution:**

- **FPN:** Top-down pathway conveying high-level semantics.
- **PANet (Default YOLOv8):** Adds a bottom-up path for low-level spatial details.
- **BiFPN:** Removes single-edge nodes, adds cross-scale residual skips and repeats blocks.

- **Weighted Feature Fusion:** Assigns learnable weights to different scale features based on importance.
- **Environmental Robustness:** Retains subtle crack cues despite factory lighting changes and shell dirt.

Feature Pyramid Architectures: FPN, PANet and BiFPN

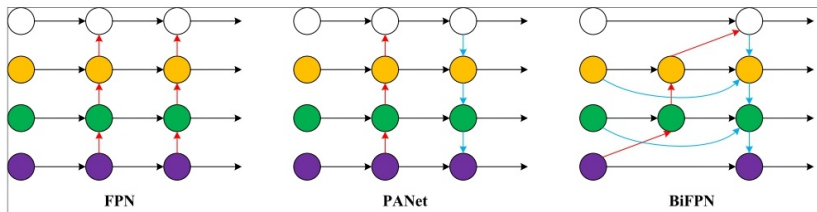


Figure: Structural comparison of FPN, PANet and BiFPN (Source: Base Paper Figure 5).

Enhancements 3 & 4: Small Target Head and WIoUv2 Loss

- **Small Target Detection Head (P2 Layer):**

- Default YOLOv8 downsamples 5 times ($/32$); smallest head (80×80) misses sub-8 px cracks.
- Added a 4th head on 160×160 feature map, directly utilizing early spatial details.

- **Wise-IoU v2 (WIoUv2) Loss Function:**

$$L_{\text{WIoUv2}} = L_{\text{IoU}}^{\gamma^*} L_{\text{WIoUv1}}, \quad \gamma > 0$$

Symbol Definitions:

- L_{WIoUv2} : Final Wise-IoU v2 bounding box regression loss.
- L_{WIoUv1} : Wise-IoU v1 loss with distance and aspect penalties.
- $L_{\text{IoU}}^{\gamma^*}$: Normalized dynamic non-monotonic focusing scaling factor.
- γ : Modulation parameter adaptively balancing gradient contributions.

Dual Verification Decision-Making Mechanism

- **Decision Rule:** An egg is classified as **damaged** if the detector outputs:
Category 2 (Cracked Egg) OR Category 3 (Crack Region)
- **Mutual Redundancy:** A cracked egg is missed **only if both** detectors fail simultaneously:
$$P(\text{Miss}_{\text{Overall}}) = P(\text{Miss}_{\text{Egg}}) \times P(\text{Miss}_{\text{Region}})$$

Symbol Definitions:

- $P(\text{Miss}_{\text{Overall}})$: Total probability of failing to identify a cracked egg.
- $P(\text{Miss}_{\text{Egg}})$: False negative rate of overall cracked egg detector.
- $P(\text{Miss}_{\text{Region}})$: False negative rate of crack region detector.

Experimental Platform and Training Setup

- **Hardware Setup:** Intel Core i5-10400F @ 2.9 GHz, NVIDIA GeForce RTX 3060 (12 GB VRAM), 32 GB DDR4 RAM.
- **Software Environment:** Windows 10, Python 3.8, PyTorch 1.13.1, CUDA 11.6.
- **Image Resolution:** 640×640 pixels.
- **Training Epochs:** 300 epochs (trained completely from scratch, batch size: 16).
- **Hyperparameters:** Initial learning rate: 0.01; random seed: 42.

Model Training and Validation Workflow

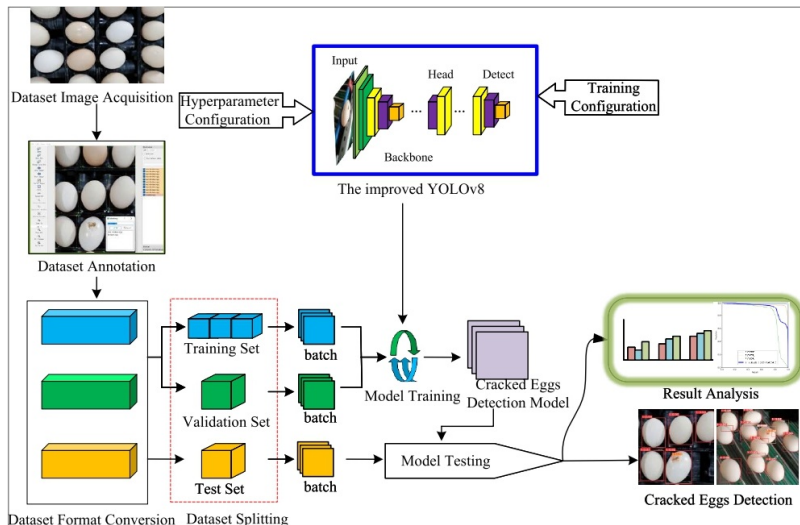


Figure: Training schematic of the cracked egg detection model (Source: Base Paper Figure 6).

Performance Evaluation Metrics

Evaluation Formulas:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$

$$\text{FNR} = \frac{FN}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN}$$

Symbol Definitions:

- TP (True Positives): Defective eggs correctly identified as cracked.
- FP (False Positives): Intact eggs incorrectly flagged as cracked (false alarms).
- TN (True Negatives): Intact eggs correctly identified as intact.
- FN (False Negatives): Cracked eggs incorrectly passed as intact (missed defects).
- **mAP@0.5**: Mean Average Precision across all classes at IoU threshold 0.5.
- **FPS**: Frames Per Second (online real-time threshold ≥ 30 FPS).

Ablation Study and Model Complexity

Table: Ablation experiment results on test dataset (Source: Base Paper Table 1)

DCNv2	BiFPN	WIoU	Small Target	Precision (%)	Recall (%)	mAP:0.5 (%)
-	-	-	-	92.2	85.3	87.7
-	-	✓	-	90.6	87.4	90.3
-	✓	-	-	89.8	83.8	88.2
-	-	-	✓	93.8	90.3	93.0
✓	-	✓	-	93.1	89.4	92.1
-	✓	✓	-	92.6	89.5	92.2
✓	✓	✓	-	93.8	89.9	92.6
✓	✓	✓	✓	94.1	90.0	92.9

- Small target head yields the largest single gain (+5.3% mAP). Full ensemble achieves highest Precision (94.1%) and balanced robustness.
- **Model Complexity (Table 2):** Parameters: 2.66×10^7 (+2.7%), FLOPs: 86.5 G (+9.3%), Model Size: 51.1 MB (+3.0%). Significant accuracy boost with minimal overhead.

Comparative Experiments with Mainstream Detectors

Table: Comparison across different object detection models (Source: Base Paper Table 3)

Model	Precision (%)	Recall (%)	mAP:0.5 (%)	FPS
Faster R-CNN (ResNet50)	68.4	80.7	78.1	17
YOLOv5	86.9	81.1	85.2	58
YOLOv8 (Baseline)	92.2	85.3	87.7	52
YOLOv11	92.7	84.3	87.3	47
Our Proposed Model	94.1	90.0	92.9	45

- Outperformed Faster R-CNN by 14.8% mAP despite Faster R-CNN using pretrained weights.
- Outperformed recent YOLOv11 by 5.6% mAP and baseline YOLOv8 by 5.2% mAP.
- At 45 FPS, detection speed easily satisfies production line requirements (≥ 30 FPS).

Laboratory Testing: Precision-Recall Curves

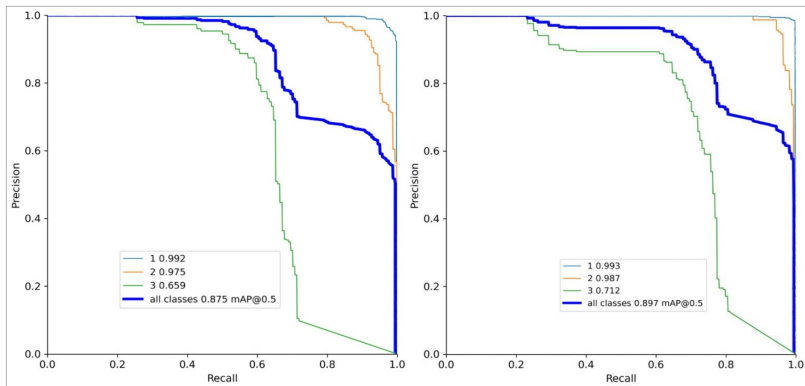


Figure: Precision-Recall curves before and after model improvement (Source: Base Paper Figure 8).

Laboratory Testing: Visual Detection Comparison



Figure: Visual detection comparison on test frames (Source: Base Paper Figure 9). Left: YOLOv8. Right: Improved YOLOv8.

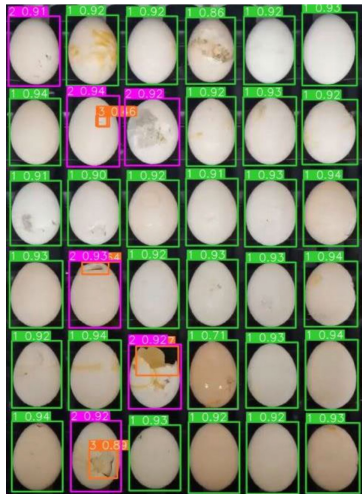
Factory Online Inspection Accuracy (FNR and FPR)

Table: Online inspection comparison on 106 production frames (Source: Base Paper Tables 5 and 6)

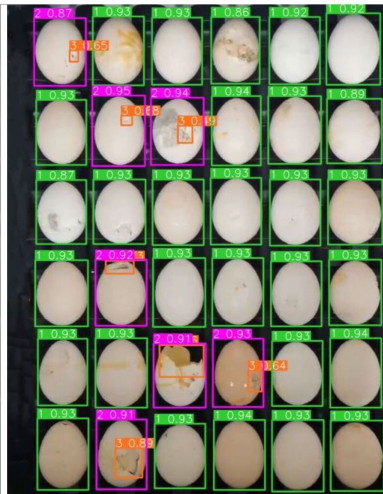
Category	Baseline YOLOv8		Proposed Improved YOLOv8	
	FNR (%)	FPR (%)	FNR (%)	FPR (%)
Class 1: Intact Egg	1.26	4.00	0.46	2.18
Class 2: Cracked Egg	10.17	3.20	5.52	1.16
Class 3: Damaged Area	22.97	0.87	19.19	0.58
Compound Dual Verification FNR	$10.17\% \times 22.97\% = 2.34\%$		$5.52\% \times 19.19\% = 1.06\%$	

- False alarm rate for intact eggs dropped from 4.00% to 2.18%, saving intact eggs.
- Compound miss rate for cracked eggs fell to 1.06%, reliably filtering defective eggs.

Production Line: Visual Detection Results



(a) YOLOv8 model detection results



(b) Improved YOLOv8 detection results

Figure: Detection effect on actual production line (Source: Base Paper Figure 10). Green: Intact, Purple: Cracked, Brown: Damaged Area.

Production Line: Detection Speed Stability

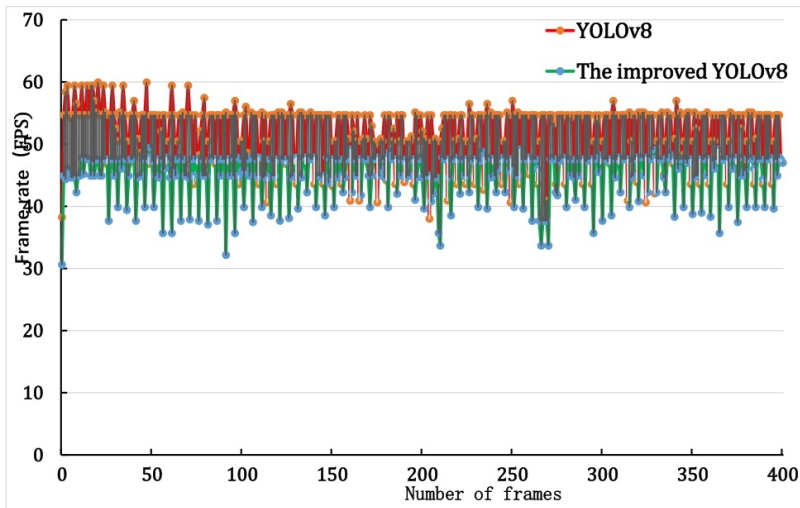


Figure: Online detection frame rate stability across 400 frames (Source: Base Paper Figure 11).

Discussion: Operational Findings and Edge Cases

- **Speed vs Accuracy Trade-off:** 14.5% speed reduction (52 to 45 FPS) gives a +5.2% mAP boost; 45 FPS well exceeds factory line demands (≥ 30 FPS).
- **Surface Adaptability:** Demonstrated high detection reliability on both unwashed farm eggs and pre-washed supermarket eggs.
- **Eggshell Varieties:** Solid shells (duck eggs) show high contrast; patterned shells (quail eggs) create background interference.
- **Reticular Micro-Cracks:** Fine net-like cracks possess low pixel areas and weak gradients, posing the primary challenge for future work.

Conclusion and Future Scope

Conclusion:

- Developed a real-time detection system combining rotational conveyance, improved YOLOv8 and dual verification.
- Integrated DCNv2, BiFPN, WIoUv2 and a P2 small target head, achieving 92.9% mAP and 45 FPS.
- Lowered compound broken egg miss rate to 1.06% on factory equipment, meeting industrial inspection criteria.

Future Scope:

- Adopt dark-box illumination setups to enhance contrast for faint net-like hairline cracks.
- Explore super-resolution and attention-guided magnification for micro-defects.
- Expand datasets to diverse poultry species and integrate internal defect detection (scattered yolk, blood spots).

References: Foundational and Survey Literature



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Thank You!

Questions and Discussion